

Examples

Verify that

$$x = At^2 + Bt + C$$

is a valid solution to

$$\ddot{x} + 4\dot{x} = 8t + 10$$

By finding $\dot{x}$ and $\ddot{x}$. Find the values of A , B and C .

$$x = At^2 + Bt + C \Rightarrow \dot{x} = 2At + B, \ddot{x} = 2A$$

$$\therefore \ddot{x} + 4\dot{x} = 2A + 8At + 4B = 8t + 10$$

$$\therefore \begin{pmatrix} 8A \\ 2A + 4B \end{pmatrix} = \begin{pmatrix} 8 \\ 10 \end{pmatrix} \Rightarrow \begin{pmatrix} A = 1 \\ B = 2 \end{pmatrix}$$

$$C = ?$$

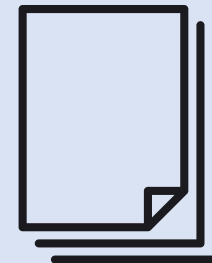
$$x = t^2 + 2t + c \text{ satisfies the equation}$$

But what about the general solution we were dealing with last time?

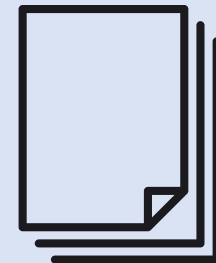
Starter

Find the general solution to

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$$



Starter



Find the general solution to

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$$

$$m^2 - 3m + 2 = 0 \Rightarrow m \in \{1, 2\}$$

$$\therefore y = Ae^x + Be^{2x}$$

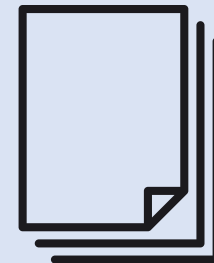
Investigating Non-Homogeneous Equations

What to do when there's a right hand side. (Hint: Don't panic)

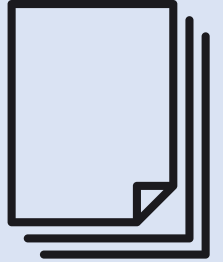
Changing the starter

What if the starter instead looked like this:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x + 7$$



Changing the starter

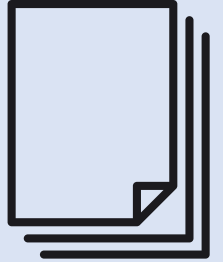


What if the starter instead looked like this:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x + 7$$

Can you verify that $y = Ax + B$ satisfies the equation, finding A and B ?

Changing the starter



What if the starter instead looked like this:

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x + 7$$

$$y = Ax + B, \frac{dy}{dx} = A, \frac{d^2y}{dx^2} = 0$$

$$\therefore \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0 - 3A + 2Ax + 2B = 2x + 7$$

$$2Ax = 2x \Rightarrow A = 1$$

$$-3A + 2B = 7 \Rightarrow B = 5$$

$\therefore y = x + 5$ satisfies the equation.

But can we go further?

What happens if we substitute $y = x + 5 + Ae^x$ into

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2x + 7$$

Since Ae^x was part of our general solution to the homogeneous.

But can we go further?

$$y = x + 5 + Ae^x, \frac{dy}{dx} = 1 + Ae^x, \frac{d^2y}{dx^2} = Ae^x$$

$$\begin{aligned} \therefore \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y &= \cancel{Ae^x} - 3 - \cancel{3Ae^x} + 2x + 10 + \cancel{2Ae^x} \\ &= 2x + 7 \end{aligned}$$

The Ae^x cancels out! That's because it makes the DE equal to 0.

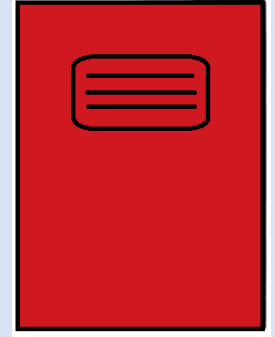
Your turn: try the whole general solution:

$$y = x + 5 + Ae^x + Be^{2x}$$

Creating our solutions

When dealing with a non-homogeneous DE, our solution comes in three parts:

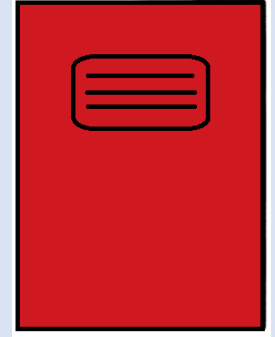
1. Particular Integral
2. Complementary Function
3. Initial Conditions



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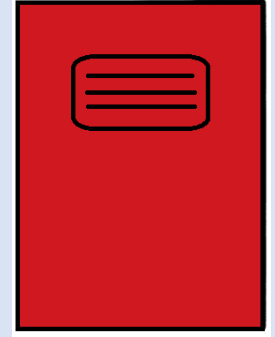
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sorts the RHS
2. Complementary Function
makes 0
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deals with unknowns



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- } General Solution



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sorts the RHS

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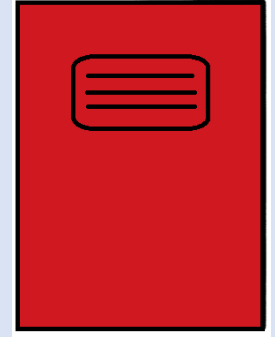
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General
Solution

Particular
Solution



Picking the trial function

But that looks so difficult! How could we ever guess the Trial Function?

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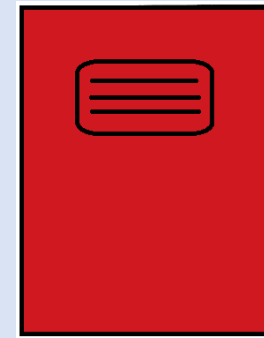
I used a linear function because the RHS was linear. Can you guess what will happen if we try a quadratic, $y = Ax^2 + Bx + C$ instead?

Particular Integrals

You'll need three “trial functions” for this course.

1. Polynomial RHS

Match the order



Particular Integrals

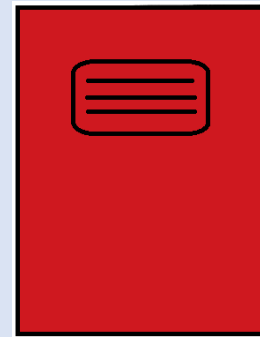
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2. Exponential RHS

Match the power



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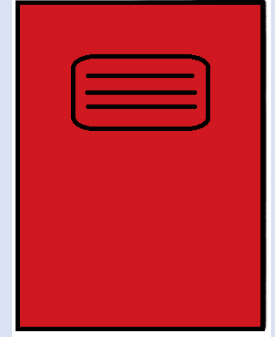
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2. Exponential RHS

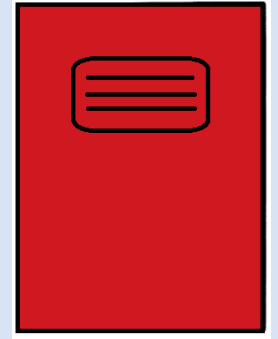
Match the power

3. Trigonometric RHS

Match the frequency



Particular Integrals



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1. Polynomial RHS

Match the order

2. Exponential RHS

Match the power

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Match the frequency

Technically, you can see a mixture.

If that happens, use a mixture too!

$$\text{EG: } \dots = 2t + e^{3t} \quad \rightarrow \quad x = At + B + Ce^{3t}$$

What would you try?

$$\ddot{x} + 3\dot{x} + x = e^{2t}$$

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What would you try?

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$$\frac{d^2P}{dq^2} - 5\frac{dP}{dq} - 24P = \sin(2q)$$

$$P = A \sin 2q + B \cos 2q$$

Trial Function Trials

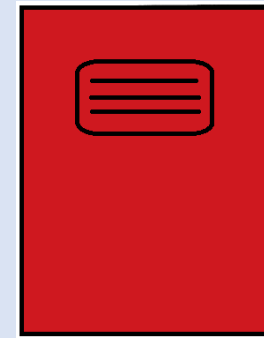
As quick as you can, write down the trial functions for:

$$\dots = e^{3x}$$

$$\dots = \cos x$$

$$\dots = 4t + 3$$

$$\dots = \cos 2x + 3 \sin 2x$$



Trial Function Trials

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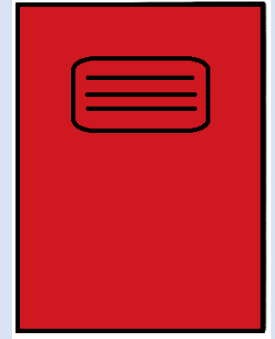
$$At + B$$

$$\dots = \cos x$$

$$A \cos x + B \sin x$$

$$\dots = \cos 2x + 3 \sin 2x$$

$$A \cos 2x + B \sin 2x$$



An exception for next time

We won't practice it today, but we can run into the repeated-roots issue with our $e^{\lambda x}$ terms in the trial function.

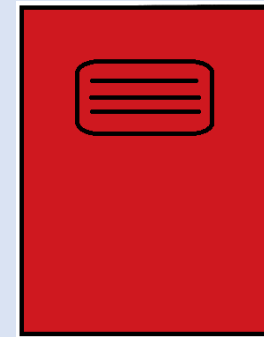
We solve it in exactly the same way, so you may end up with $Ae^x + Bxe^x + Cx^2e^x$ if you're **really** unlucky.

In general, you want to work out the Complementary Function first, and then a trial function afterwards. That way you can do it in one go.

Finding the Particular Integral

Find the Particular Integral for:

$$\ddot{x} - 5\dot{x} + x = e^{2t}$$



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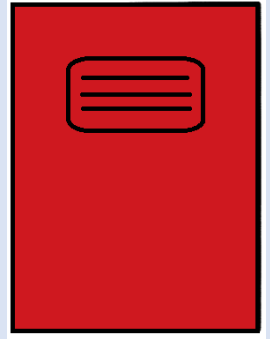
$$\ddot{x} - 5\dot{x} + x = e^{2t}$$

$$\text{TF : } x = Ae^{2t} \Rightarrow \dot{x} = 2Ae^{2t}, \ddot{x} = 4Ae^{2t}$$

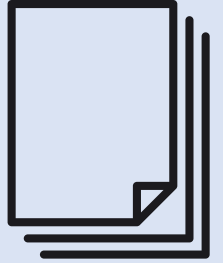
$$\therefore \ddot{x} - 5\dot{x} + x = 4Ae^{2t} - 10Ae^{2t} + Ae^{2t} = -5Ae^{2t}$$

$$-5Ae^{2t} = e^{2t} \Rightarrow A = -\frac{1}{5}$$

$$\therefore \text{PI : } x = -\frac{1}{5}e^{2t}$$



Practice



Find the Particular Integrals for the following functions:

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 3x - 2$$

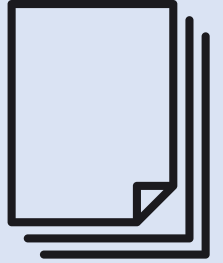
$$\ddot{x} - 2\dot{x} + 7x = 4e^t$$

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 4\cos x$$

$$\frac{d^2P}{dq^2} - \frac{dP}{dq} - 2P = 5\sin 2q$$

$$\ddot{x} - 7\dot{x} + 2x = t^2 - 7t + 2$$

Practice



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$$y = x - 2$$

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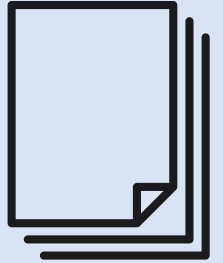
$$y = -2\sin x$$

$$\ddot{x} - 7\dot{x} + 2x = t^2 - 7t + 2$$

$$\ddot{x} - 2\dot{x} + 7x = 4e^t$$

$$x = \frac{2}{3}e^t$$

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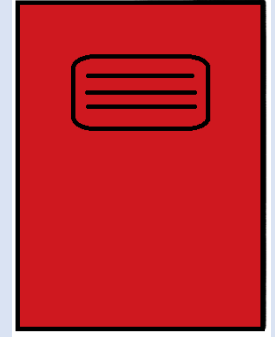
$$P = \cos 2q - 2\sin 2q$$

$$x = \frac{1}{2}t^2 + \frac{1}{2}$$

The General Solution

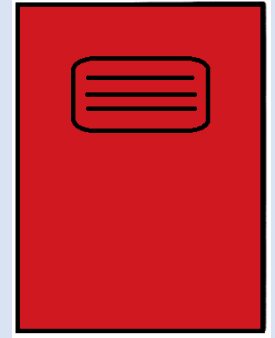
The General Solution is formed by adding the Particular Integral and Complementary Functions together.

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The General Solution

The General Solution is formed by adding the Particular Integral and Complementary Functions together.



$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = 3x - 2$$

$$\text{CF : } m^2 + 4m + 3 = 0$$

$$\Rightarrow m \in \{-1, -3\}$$

$$\Rightarrow y = Ae^{-x} + Be^{-3x}$$

$$\text{PI : try } y = Ax + B$$

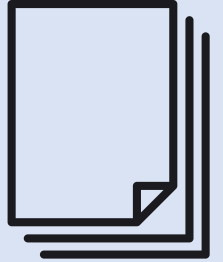
$$\Rightarrow dy = A, d^2y = 0$$

$$\therefore 0 + 4A + 3Ax + 3B = 3x - 2$$

$$\therefore A = 1, B = -2$$

$$\therefore y = x - 2 + Ae^{-x} + Be^{-3x}$$

Practice



Find the General Solutions for the following functions:

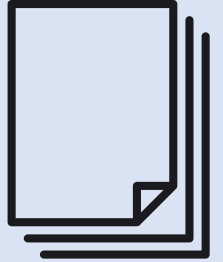
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Practice



Find the General Solutions for the following functions:

$$\ddot{x} - 2\dot{x} + 7x = 4e^t$$

$$x = \frac{2}{3}e^t + e^t(A \sin \sqrt{6}t + B \cos \sqrt{6}t)$$

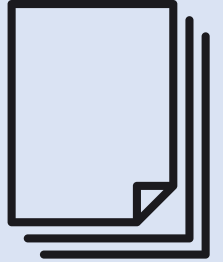
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$$\ddot{x} - 7\dot{x} + 2x = t^2 - 7t + 2$$

$$x = \frac{1}{2}t^2 + \frac{1}{2} + Ae^{\frac{7+\sqrt{41}}{2}t} + Be^{\frac{7-\sqrt{41}}{2}t}$$